SOVEREIGN BOND CONVENIENCE YIELDS AS FINANCIAL REPRESSION: PRICE AND REAL EFFECTS

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Abstract. Governments induce demand on their own bonds through financial regulatory policy. This induced demand relaxes their budget constraint. If fiscal policy plays some role in pinning down the price level, changes in financial regulation might influence inflation and real variables. This dissertation explores the plausibility of this channel by building a simple micro-founded model of a reduced form regulatory-constrained financial intermediary in an active-fiscal environment with price rigidities. The model provides a link between financial regulation and empirically documented sovereign bond pricing puzzles. Calibrated to UK data, the model shows how changes in regulatory policy move prices, and the extent to which the extra fiscal space generated from these "repression rents" substitutes for fiscal surpluses, keeping inflation constant.

1. Introduction

Many advanced economies face a common fiscal challenge. Population ageing is increasing expenditure pressures through pensions, healthcare, and other age-related transfers, while political constraints make large-scale fiscal adjustment increasingly difficult. As public debt accumulates and demographic trends continue to deteriorate, governments may face growing incentives to seek alternative margins of adjustment beyond conventional taxation and expenditure reform.

Recent work has revived interest in precisely this possibility. Jeanne (2025) argues that persistent fiscal deadlock can create incentives for increasingly interventionist forms of financial repression, while Reis (2025) highlights the range of policy instruments through which governments can influence sovereign financing conditions. At the same time, a growing literature documents a "Sovereign Debt Valuation Puzzle" (Jiang et al, 2024), as government debt frequently trades at prices difficult to reconcile with frictionless asset-pricing benchmarks, suggesting that institutional distortions, captive demand, and regulatory constraints may play an important role in sovereign valuation.

This dissertation asks: **Can financial regulatory policy affect inflation and real economic activity?**

More specifically: **Is there a way to write a model where there is a transmission channel from financial regulation to prices and real variables? If so, does the evidence and literature support the assumptions and conditions on such a model?**

For a channel through which the effect may operate to exist, two things need to hold together: Financial regulatory policy relaxing the government's budget constraint, and the government's budget constraint playing a role in determining inflation and real variables.

In this dissertation, I write a micro-founded model borrowing from Intermediary Asset Pricing (IAP), the Fiscal Theory of the Price Level (FTPL), and New Keynesian (NK) rigidities, which produces the argued effect: financial regulatory policy affects prices and real variables.

The effect of regulation on the government's budget constraint is shown to be true within the model as an equilibrium accounting relation, whereas the effect on prices is shown to depend on equilibrium selection and determinacy conditions.

The model is solved under a simple calibration. I present and discuss impulse responses to a regulatory shock, the trade-off between repression and surpluses, and alternative regimes and calibrations.

2. Literature Review

A debtor cannot rely forever on rolling over its obligations. Its outstanding debt must ultimately be paid for by future income, by a reduction in the value of what it owes, or by creditors accepting lower-than-normal returns. Reis (2025) states this as:

(1) Debt = EPV(net income) + fall in market value of the debt + EPV(discounts on debt returns).

Applied to government debt, the conventional narrative regarding the large stock of outstanding debt by many governments is that net income (primary surpluses) are an unconstrained residual: whatever backing the debt requires, the fiscal authority will eventually be able to supply it through some future combination of higher taxes and lower expenditure. The premise of this dissertation is that this residual is not freely available in many countries, particularly Western Europe. With electorates increasingly dominated by retirees and near-retirees, and with the demographic pressures documented by Fernández-Villaverde (2024) and the European Commission (2024) raising age-related expenditure faster than revenue, the surpluses required to stabilize the sovereign balance sheet come to lie above what the political system can credibly deliver over the relevant horizon.

The devaluation margin is constrained as well. Outright default is extremely costly, and real devaluation through inflation is constrained by institutions rather than electorates: three decades of central-bank independence with explicit anti-inflation mandates make it hard to believe that XXIst century institutional arrangements would allow for an inflationary resolution to the problem (Bernanke, 2022).

Two of the three margins of adjustment are obstructed at once. Thus, the adjustment that the identity requires must fall on the remaining margin, a persistent compression of the discount rates applied to sovereign liabilities. This has happened before. In Great Britain, the very large public debts left by the Second World War were paid off largely along this margin (Reinhart & Sbrancia, 2015), though in more explicitly repressive Bretton Woods era financial regulatory environment than what exists today, where capital flows are mostly free and distortionary regulations are justified on other grounds.

Imposing similar constraints on the margins of adjustment, Jeanne (2025) derives financial repression as an endogenous outcome of a sovereign minimizing the welfare cost of servicing a debt it cannot stabilize through ordinary means. In a Ramsey problem closely related to the optimal-repression analysis of Chari, Dovis, and Kehoe (2020), the sovereign reaches for quasi-fiscal revenue from the financial sector only once ordinary taxation, more efficient at the margin, has been exhausted up to its political ceiling; repression is therefore the residual instrument, the one deployed last because it is the costliest, and it appears precisely in the configuration this dissertation studies. Additionally, he provides a lexicographic ordering of specific policy instruments of financial repression that a government under fiscal deadlock may reach for, providing a temporal logic to endogenous financial repression levers.

Reis (2025) supplies a cross-sectional map of the instruments of financial repression, operating on different asset classes and maturities. The idea is that people hold assets not only because of the monetary returns they earn from them, but also because of the direct benefits derived from holding. The most trivial way to illustrate the logic is that people hold cash at zero yield, because it provides the services of anonymity and liquidity. The benefit of holding instruments can be a function of the regulatory environment the holders live in. Liquidity provides a benefit in and of itself, but Basel III regulation mandating banks to hold High Quality Liquid Assets increases that baseline. On the long end, regulation imposed on insurers and pension funds making them match the duration of their assets to the duration of their liabilities increase the benefit of holding long-maturity bonds, and hence the demand for them.

Even though the primary objective of regulation may very well be orthogonal to fiscal stability concerns, the price distortion nevertheless relaxes the government's budget constraint, as it generates a wedge between the price that government sells its issued debt for, and the price that they would get if the regulations did not exist. Because fiscal reform, inflation, and default are all very costly, while discount compression operates through the indirect and politically low-salience machinery of prudential regulation, it's a reasonable belief (or at least, sufficiently reasonable to warrant exploration) that this will be used as a margin of fiscal adjustment.

Thus, I have narrowed down onto a single, well-defined term of a recognized taxonomy: the rent that sovereign extracts through regulation-induced demand for its liabilities driving the price above what it would be under no regulatory requirements.

This has already been extensively documented in the macro-finance literature. Krishnamurthy and Vissing-Jorgensen (2012) provide the canonical estimates of safety and liquidity premia on US Treasuries, and Greenwood and Hanson (2013) develop closely related evidence on the pricing of and demand for safe assets. More recently, Jiang, Lustig, Van Nieuwerburgh, and Xiaolan (2024) sharpen the same observation into a named "Sovereign Debt Valuation Puzzle", showing that the market value of US federal debt exceeds the present discounted value of projected surpluses by a wide margin they cannot reconcile with frictionless benchmarks. The standard explanation is that these bonds provide priced services on top of their cash flows. It's "convenient" to hold them (for safety, liquidity, pledgeability, duration, or other reasons) and as such there is a "convenience yield" that covers the gap. This empirical literature is deliberately agnostic about mechanism: the wedge is measured as a reduced-form residual, with no structural account of why the "convenience" exists.

The intermediary asset-pricing literature is borrowed to fill that gap by replacing the frictionless representative household with a constrained financial intermediary as the marginal investor, so that an observed price reflects the shadow value of a binding constraints rather than an unconstrained stochastic discount factor. He and Krishnamurthy (2013) and Brunnermeier and Sannikov (2014) develop the canonical versions in which intermediary net worth and leverage drive risk premia, Gertler and Karadi (2011) provide the DSGE template that embeds a constrained intermediary in a New Keynesian environment, and He, Kelly, and Manela (2017) and Adrian, Etula, and Muir (2014) show empirically that intermediary balance-sheet variables price a broad cross-section of assets.

For modeling purposes, I focus specifically on the distortion in long-maturity bond markets generated by duration-matching requirements, a common feature of prudential regulation across many countries. Institutions with long-dated liabilities (pension funds and life insurers above all) have a cap on the gap between the duration of their assets and that of their liabilities. The logic is immunizing against interest-rate risk, requiring them to hold long-duration assets to match their long duration obligations. It's often the case that the corporate bond market does not supply enough fixed-income, long-maturity debt to satisfy these regulations, so the financial intermediaries are forced to hold long-term government bonds. The regulation therefore manufactures price-inelastic demand, bidding the price of long-dated sovereign debt above what an unconstrained investor would pay.

This rationalizes the wedge directly: when regulatory mandates force institutions such as pension funds and insurers to hold long-maturity government debt regardless of its frictionless valuation, the observed price is bid above what an unconstrained investor would pay, and the gap is the shadow value of the binding constraint (He & Krishnamurthy, 2013; Gertler & Karadi, 2011). The specific friction relevant here is the demand for duration, for which Vayanos and Vila (2021) provide the preferred-habitat foundation and Greenwood and Vayanos (2014) the supporting evidence on maturity-structure supply and bond returns; the mechanism is not merely theoretical, since the United Kingdom's 2004 pension reform manufactured exactly the price-inelastic long-end demand that Greenwood and Vayanos (2010) show compressed long-end yields in the way preferred-habitat logic predicts (Vayanos & Vila, 2021).

The wedge has now been described twice, as a measured convenience yield with no stated source and as the shadow price of an intermediary's binding constraint. My contribution follows a commonly made observation across these two separate literatures: the service flow that the convenience-yield literature attributes to non-pecuniary benefits is itself, to a substantial degree, manufactured by regulation, since liquidity coverage ratios, risk-weighted capital requirements, collateral-eligibility rules, and duration-matching mandates all generate institutional demand that would not exist in a frictionless market.

If the convenience yield is the price of a benefit, and the benefit is produced by regulation, then the convenience yield and the repression rent are not distinct objects but the same wedge read through different lenses, the one exogenous demand for safety and the other the endogenous outcome of policy.

Whether that enlarged backing affects the price level is another matter. An extensive research program has developed over the last few decades concerning the role of the government's budget constraint in determining the price level. There are two insights driving this literature. First, that the ultimate nominal anchor of an economy (specially in a fiat-money system) is the requirement by governments for taxes to be paid in its currency; second, that real government debt is a priced asset, and as such, is valued as the discounted stream of future cash flows, or in this case, future primary surpluses:

$$\frac{B}{P} = \sum_{j=0}^{\infty} \beta^j s_{t+j} \tag{2}$$

where B is the nominal stock of outstanding bonds, P is the price level (so that B/P is the real stock of outstanding bonds), s are surpluses and β a discount factor.

This equation is not controversial. It follows directly from the government's per-period budget constraint (which is an accounting identity) once a terminal condition is imposed. The controversy is over the equilibrium closure: which object is allowed to adjust so that the restriction holds. Canonical NK models typically impose a passive fiscal closure, building in a Ricardian fiscal backing condition and removing the fiscal determination of the price level by assumption.

Sargent and Wallace (1981) showed through some unpleasant arithmetic that a central bank facing a fiscal authority committed to persistent deficits cannot deliver low inflation indefinitely, and that the long-run price level cannot be determined by the monetary rule alone.

If the surplus path is exogenous (active fiscal authority), given a discount factor, the (predetermined) stock of nominal debt, the price level remains the only variable that can move to equilibrate (2)

That is settled by the monetary-fiscal regime, and the organizing device is the active/passive classification of Leeper (1991), which crosses the stance of monetary policy with that of fiscal policy and shows that a determinate equilibrium requires exactly one active authority, pursuing its own target without regard to debt, and one passive authority, adjusting to keep the intertemporal budget satisfied. This produces four configurations, and only the two diagonal ones are determinate. If both the fiscal and the monetary authorities are pursuing their objectives actively; the system is not determinate. If both are passive, the system is underdetermined.

	Active fiscal (exogenous s)	Passive fiscal (Ricardian Rule)
Active money ($\phi_\pi > 1$)	no equilibrium	determinate
Passive money ($\phi_\pi < 1$)	determinate	under-determined

active money with passive fiscal, the conventional Ricardian closure in which the central bank pins inflation and the treasury validates it, and passive money with active fiscal, the fiscal-dominance closure that the FTPL articulates and in which the price level adjusts to clear the valuation identity (Leeper, 1991; Woodford, 1995). The entire distinction turns on a single inequality on the monetary-policy coefficient, so that the active/passive choice is joint rather than separate, a property emphasized by Leeper (1991) and made central to the synthesis of Cochrane (2023).

I ought to highlight that the question of which of these two worlds we actually live in is not settled by the data and remains an open question in the literature. The regimes differ only in off-equilibrium responses that are never observed, and they can generate identical in-sample paths for inflation, debt, surpluses, and rates, an observational-equivalence problem treated candidly by Cochrane (2023). The choice of regime therefore rests on a judgment about the policy environment rather than on a test, and the case for fiscal dominance in the present setting rests on three grounds.

The first follows from Jeanne's (2025) insight that median-voter pressures in ageing democracies make Ricardian closure arguments hard to believe in relevant horizons, and as such surpluses are bounded from above by a political ceiling, the fiscal authority is active by necessity rather than choice, and a coherent equilibrium then requires monetary policy to be passive and the price level to do the equilibrating work the surplus cannot.

The second ground is theoretical and concerns determinacy itself. Determinacy here refers to the existence and uniqueness of a bounded equilibrium in a linearized rational-expectations model, which to exist requires the number of unstable roots to equal the number of forward-looking jump variables (Blanchard and Kahn, 1980). Too few explosive roots leave a continuum of bounded paths (indeterminacy) and too many leave none. Sims (2002) recasts the same condition in a form that does not require the model to be partitioned by hand into predetermined and jump blocks (the method used in this dissertation). The standard case for the Ricardian closure rests on the claim that the Taylor principle pins down the price level, but Benhabib, Schmitt-Grohé, and Uribe (2001) show that an active Taylor rule is consistent with a continuum of bounded trajectories that satisfy the Euler equation, the rule, and transversality while drifting from the target toward a second, deflationary steady state. The resolution they and subsequent authors identify is a fiscal backstop. Cochrane (2023) extends this reading into a general argument for explicitly modeling fiscal dynamics instead of just implicitly assuming Ricardian closure, as it's needed as an equilibrium-selection mechanism in order to complete the system.

The third ground is empirical: the post-covid inflation reopened the question of whether advanced-economy price levels are governed by monetary or fiscal forces. Bianchi and Melosi (2017, 2022) develop regime-switching frameworks in which an economy slips between configurations as fiscal conditions evolve, arguing that a fiscal component is needed to account for the inflation of the early 2020s.

There is also empirical evidence to support the idea that financial repression substitutes for fiscal surpluses and can help keep inflation under control even under severe fiscal dominance. The most standard objection to the FTPL is the case of Japan, a country that has been running deficits for decades and whose stock of public debt is over two times its output, yet keeps inflation low. Chien, Cole, and Lustig (2023) show that the consolidated Japanese public sector has been borrowing at severely repressed yields, taking advantage of a repressed base of savers. They estimate that the return on the consolidated public sector's debt is at least 180 basis points per annum below what the risk on its balance sheet would warrant: a "government debt valuation wedge" they attribute to financial repression allows the public sector to harvest an excess return averaging 2.28% of GDP per year over 1998–2023, and 6.25% of GDP per year over the 2013–2023 decade. This dissertation can be read as a (very reduced form) structural counterpart to the mechanism they describe.

My contribution is an expression of Cochrane's standard backing object, where the regulation-induced price distortion generates a per-period rent that is a genuine source of fiscal backing alongside the surplus, granted fiscal dominance, this becomes the equation that selects the price level. The effect of inflation dynamics on real variables will follow from standard New-Keynesian specification (Galí, 2015; Calvo, 1983).

3. The Model

Time is discrete and infinite, $t = 0, 1, 2, \ldots$, one period a quarter. There is a single non-storable good. The economy contains a representative household, a continuum of monopolistically competitive firms $i \in [0, 1]$, a treasury, and a monetary authority. Expectations are rational; $\mathbb{E}_t$ is the date-t conditional expectation. The price level is P_t, gross inflation $\pi_t \equiv P_t/P_{t-1}$.

The government issues two nominal bonds.

- A *short bond*: a one-period claim; one unit bought at t at real price Q_t^S pays one nominal unit at $t + 1$.
- A *long bond*: a geometric perpetuity paying nominal coupons $1, \delta, \delta^2, \ldots$ at $t + 1, t + 2, \ldots$, $\delta \in [0, 1)$. A unit issued k periods ago is a perfect substitute, scaled by δ^k, for a fresh unit, so the maturity structure collapses to a single price Q_t^L and stock b_t^L; the real value of one previously issued long bond is $1 + \delta Q_t^L$.

Assumption 1 (Two arms, one discount factor)**.** The household pools consumption across its arms, so a single marginal utility (hence a single stochastic discount factor) prices both bonds. The captive arm holds the long bond subject to a binding coverage constraint; the unconstrained arm holds the short bond freely.

Assumption 2 (Segmentation)**.** The household cannot take an offsetting unconstrained position in the long bond; it is priced by the captive arm's constrained optimality condition, and no position exists to arbitrage away the wedge.

Assumption 3 (Binding regime)**.** The analysis is conducted near a steady state at which the coverage constraint binds with strictly positive multiplier; the equilibrium imposes the binding branch.

The representative household's per-period utility is additively separable in consumption c_t and labor N_t,

$$U(c_t, N_t) = \frac{c_t^{1-\sigma}}{1-\sigma} - \psi\, \frac{N_t^{1+\varphi}}{1+\varphi}, \qquad \sigma > 0,\ \varphi \geq 0,\ \psi > 0, \tag{3}$$

and they maximize $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t, N_t)$, $\beta \in (0, 1)$. Because $\partial U/\partial c_t = c_t^{-\sigma}$ does not involve N_t, the consumption Euler equation and the stochastic discount factor are independent of the labor margin; labor enters only through firms' marginal cost.

The household holds real balances b_t^S, b_t^L and faces, each period, the real flow budget constraint

$$c_t + Q_t^S b_t^S + Q_t^L b_t^L = y_t - \tau_t + \frac{b_{t-1}^S}{\pi_t} + \frac{(1 + \delta Q_t^L) b_{t-1}^L}{\pi_t}, \tag{4}$$

with τ_t a lump-sum tax, and the regulatory coverage constraint on the captive arm

$$Q_t^L b_t^L \geq m_t \equiv \lambda_t \bar{V}, \tag{5}$$

which requires the real market value of long-bond holdings to be at least a regulated fraction λ_t (the policy instrument) of a fixed liability pool $\bar{V}$.

Attaching multipliers $\beta^t \Lambda_t$ to (4) and $\beta^t \nu_t$ ($\nu_t \geq 0$) to (5), the household's first-order conditions are derived term by term in Appendix 1 . The consumption FOC gives $\Lambda_t = c_t^{-\sigma}$, which defines the stochastic discount factor

$$M_{t+1} \equiv \beta\, \frac{\Lambda_{t+1}}{\Lambda_t} = \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\sigma}. \tag{6}$$

The short-bond FOC gives the short Euler equation

$$Q_t^S = \mathbb{E}_t\left[\frac{M_{t+1}}{\pi_{t+1}}\right], \qquad 1 + i_t \equiv \frac{1}{Q_t^S}. \tag{7}$$

The long-bond FOC, with the normalized multiplier $\tilde{\nu}_t \equiv \nu_t/\Lambda_t \geq 0$, gives the long Euler equation

$$Q_t^L\,(1 - \tilde{\nu}_t) = \mathbb{E}_t\left[M_{t+1}\frac{1 + \delta Q_{t+1}^L}{\pi_{t+1}}\right]. \tag{8}$$

The right side is the value of the bond's cash flows under the household's kernel; the left is the market price scaled by $1-\tilde{\nu}_t$. When the mandate binds ($\tilde{\nu}_t > 0$) the market price exceeds the kernel value: regulation forces the captive arm to hold the bond above the price an unconstrained investor would pay. The Kuhn–Tucker conditions for (5) are

$$\nu_t \geq 0, \qquad Q_t^L b_t^L - m_t \geq 0, \qquad \nu_t\big(Q_t^L b_t^L - m_t\big) = 0. \tag{9}$$

Under Assumption 3 the constraint binds, $Q_t^L b_t^L = m_t$ and $\tilde{\nu}_t > 0$; if it were slack, $\tilde{\nu}_t = 0$ and (8) would reduce to an ordinary perpetuity Euler equation.

Definition 1 (Fundamental price)**.** The fundamental long-bond price is the price under no constraints, i.e. (8) with $\tilde{\nu}_t \equiv 0$:

$$Q_t^{L,f} = \mathbb{E}_t\left[M_{t+1}\frac{1 + \delta Q_{t+1}^{L,f}}{\pi_{t+1}}\right]. \tag{10}$$

Definition 2 (Wedge)**.** $1 + \omega_t \equiv Q_t^L/Q_t^{L,f}$, with $\omega_t \geq 0$.

In the binding regime, $\tilde{\nu}_t > 0$ implies $Q_t^L > Q_t^{L,f}$, hence $\omega_t > 0$.

Definition 3 (Repression rent)**.** $R_t \equiv \tilde{\nu}_t\, Q_t^L b_t^L$.

By (8) the kernel value of the long bond's cash flows is $Q_t^L(1-\tilde{\nu}_t)$ while its market price is Q_t^L; the per-unit gap is $\tilde{\nu}_t Q_t^L$, and multiplied by the stock b_t^L it is the resource transfer R_t from the captive arm to the issuer.

Each period the government finances itself from two sources: lump-sum taxation τ_t and the real revenue raised by issuing new debt, $Q_t^S b_t^S + Q_t^L b_t^L$, and deploys those resources to redeeming the debt that matures into period t and paying for government spending g_t. Equating sources and uses,

$$\underbrace{\tau_t \;+\; Q_t^S b_t^S + Q_t^L b_t^L}_{\text{taxation}\,+\,\text{new issuance}} \;=\; \underbrace{\frac{b_{t-1}^S + (1+\delta Q_t^L)\, b_{t-1}^L}{\pi_t}}_{\text{maturing bonds}} \;+\; \underbrace{g_t}_{\text{spending}}.$$

The long bond is a geometric perpetuity paying nominal coupons $1, \delta, \delta^2, \ldots$ from $t+1$ onward. A unit issued k periods ago is a perfect substitute, scaled by δ^k, for a freshly issued unit, so every vintage of long debt collapses into a single stock b_t^L and a single price Q_t^L. Consequently one previously issued long bond carried into period t is worth $1+\delta Q_t^L$ in real terms (the coupon of 1 plus the continuation value δQ_t^L of the remaining claim). This is why the maturing long-bond term enters with the factor $(1 + \delta Q_t^L)$.

Move g_t alongside τ_t to form the primary surplus $s_t \equiv \tau_t - g_t$ and group the maturing-debt term as the state variable. Define the real value of debt carried into t as v_t and the market value of debt issued at t as $\tilde{v}_t$:

$$\underbrace{\frac{b_{t-1}^S + (1+\delta Q_t^L)\, b_{t-1}^L}{\pi_t}}_{\equiv v_t} = \underbrace{(\tau_t - g_t)}_{= s_t} + \underbrace{Q_t^S b_t^S + Q_t^L b_t^L}_{\equiv \tilde{v}_t}, \qquad s_t \equiv \tau_t - g_t. \tag{11}$$

That is, $v_t = s_t + \tilde{v}_t$: inherited real debt equals the surplus plus the market value of new issuance.

From this point on, fiscal policy is specified in terms of the primary surplus s_t, not by separate rules for taxes and government purchases. Government purchases g_t enter the resource constraint, while lump-sum taxes adjust residually according to $\tau_t = s_t + g_t$. Thus, a movement in g_t is not, by itself, a deficit-financed spending shock unless it is accompanied by a movement in s_t.

Evaluating the definition of v at $t+1$,

$$v_{t+1} = \frac{b_t^S + (1 + \delta Q_{t+1}^L)\, b_t^L}{\pi_{t+1}}. \tag{12}$$

Multiply (12) by the stochastic discount factor M_{t+1} and take the date-t expectation. Because b_t^S and b_t^L are chosen at t, they factor out of $\mathbb{E}_t[\cdot]$ exactly, leaving the two bond-pricing expectations intact:

$$\mathbb{E}_t[M_{t+1} v_{t+1}] = b_t^S \underbrace{\mathbb{E}_t\left[\frac{M_{t+1}}{\pi_{t+1}}\right]}_{=Q_t^S} + b_t^L \underbrace{\mathbb{E}_t\left[\frac{M_{t+1}(1 + \delta Q_{t+1}^L)}{\pi_{t+1}}\right]}_{=Q_t^L(1-\tilde{\nu}_t)}, \tag{13}$$

the first bracket being the short-bond Euler equation (7) and the second the long-bond Euler equation (8).

Substituting the two prices and using $\tilde{v}_t = Q_t^S b_t^S + Q_t^L b_t^L$ together with the rent definition $R_t \equiv \tilde{\nu}_t Q_t^L b_t^L$,

$$\mathbb{E}_t[M_{t+1} v_{t+1}] = Q_t^S b_t^S + Q_t^L b_t^L (1 - \tilde{\nu}_t) = \underbrace{Q_t^S b_t^S + Q_t^L b_t^L}_{=\tilde{v}_t} - \underbrace{\tilde{\nu}_t Q_t^L b_t^L}_{=R_t} = \tilde{v}_t - R_t. \tag{14}$$

The wedge enters the long-bond Euler multiplicatively, as the scalar $(1-\tilde{\nu}_t)$ on a single predetermined quantity; reconstructing the market value $\tilde{v}_t$ therefore leaves the constraint's entire footprint as the clean additive term $-R_t$, with no cross term.

Rearranging (14) expresses $\tilde{v}_t$ as the rent plus the discounted continuation value of debt:

$$\tilde{v}_t = R_t + \mathbb{E}_t[M_{t+1} v_{t+1}].$$

The value the government raises by issuing today equals the repression rent it extracts this period plus the present value of the debt it carries forward. The rent is the value raised by today's issuance net of the present value of the obligation it creates, as $R_t = \tilde{v}_t - \mathbb{E}_t[M_{t+1} v_{t+1}]$. Because the continuation value v_{t+1} carries the inherited long bonds at their revalued price $(1 + \delta Q_{t+1}^L)$, the revaluation of outstanding long debt is already deducted inside $\mathbb{E}_t[M_{t+1} v_{t+1}]$; the rent is what survives that netting, not a gross issuance receipt. Under fair pricing ($\tilde{\nu}_t = 0$) the two sides coincide and $R_t = 0$, so the wedge is precisely what opens the gap.

Combining $\tilde{v}_t = R_t + \mathbb{E}_t[M_{t+1} v_{t+1}]$ with $v_t = s_t + \tilde{v}_t$ from (11) gives the recursive augmented identity

$$v_t = s_t + R_t + \mathbb{E}_t[M_{t+1} v_{t+1}]. \tag{15}$$

Iterating (15) forward T times, discounting, and using the law of iterated expectations, with the convention $\prod_{k=1}^{0} M_{t+k} \equiv 1$, we obtain:

$$v_t = \mathbb{E}_t \sum_{j=0}^{T} \left(\prod_{k=1}^{j} M_{t+k}\right)\left(s_{t+j} + R_{t+j}\right) \; + \; \mathbb{E}_t\left[\left(\prod_{k=1}^{T+1} M_{t+k}\right) v_{t+T+1}\right].$$

Imposing the no-bubble transversality condition

$$\lim_{T\to\infty} \mathbb{E}_t\left[\left(\prod_{k=1}^{T+1} M_{t+k}\right) v_{t+T+1}\right] = 0 \tag{16}$$

makes the terminal term vanish, leaving the augmented valuation identity

$$v_t = \mathbb{E}_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^{j} M_{t+k} \right) \big(s_{t+j} + R_{t+j} \big). \tag{17}$$

Relative to the standard surplus-only identity, the backing is enlarged by the present value of rents: repression is an active margin of adjustment, not a distortion sitting on the system's accounting margin.

The price level enters through v_t. Writing the predetermined nominal stocks $b^j_{t-1} = B^j_{t-1}/P_{t-1}$ and $\pi_t = P_t/P_{t-1}$, the definition of v_t in (11) becomes $v_t = B_t/P_t$, where $B_t \equiv B^S_{t-1} + (1 + \delta Q^L_t) B^L_{t-1}$ is the nominal value of inherited claims. Substituting into (17),

$$\frac{B_t}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^{j} M_{t+k} \right) \big(s_{t+j} + R_{t+j} \big), \tag{18}$$

A tighter mandate raises the R_{t+j}, raises the denominator, and, holding the numerator B_t fixed (thereby abstracting from the contemporaneous revaluation of inherited long debt through Q^L_t, the standard Cochrane cross-steady-state reading I adopt for the propositions below and relax on the dynamic transition), calls for a lower P_t: repression substitutes for inflation on the equilibrating margin.

Relative to the standard Cochrane (2023) identity, the backing is enlarged by the present value of rents. I call this the "augmented FTPL identity".

The regulatory instrument λ_t enters the model at a single point: the coverage constraint (5) fixes the captive bucket at $Q^L_t b^L_t = \lambda_t \bar{V}$. The rent the binding mandate extracts is therefore

$$R_t = \tilde{\nu}_t \, Q^L_t b^L_t = \tilde{\nu}_t \, \lambda_t \bar{V}. \tag{19}$$

Proposition 1 (Regulation relaxes the budget constraint)**.** *In the binding regime ($\tilde{\nu} > 0$), the repression rent is strictly increasing in the regulatory instrument:*

$$\frac{\partial R}{\partial \lambda} = \tilde{\nu} \, \bar{V} > 0. \tag{20}$$

Proof. Differentiate $R = \tilde{\nu} \, \lambda \, \bar{V}$ from (19) holding $\tilde{\nu}$ fixed; since $\bar{V} > 0$ and $\tilde{\nu} > 0$ in the binding regime, $\tilde{\nu}\bar{V} > 0$. Across steady states this is the whole effect: $\tilde{\nu}$ is pinned by the calibrated wedge ω through (32), independent of λ, so $d\tilde{\nu}/d\lambda = 0$ and the total derivative coincides with the partial. □

The partial is an accounting property: it requires neither nominal rigidity nor a particular monetary–fiscal regime. On the dynamic transition the wedge itself responds ($\hat{\omega}_0 > 0$), so the impact rent response exceeds the partial; Appendix 5 reports this as the impact elasticity $\hat{R}_0/\hat{\lambda}_0 \approx 116$ and verifies its sign and determinacy across the calibration. Whether the additional backing is absorbed by the price level or by the surplus is settled by the determinacy conditions of the next section, on which this proposition is silent.

Additionally,

Proposition 2 (Surplus substitution)**.** *Holding the price level fixed, the rent that the mandate generates substitutes one-for-one for the primary surplus in the fiscal backing of the debt:*

$$\left. \frac{ds}{d\lambda} \right|_P = -\, \tilde{\nu} \bar{V} < 0. \tag{21}$$

Proof. Evaluate the augmented identity (15) in steady state ($M = \beta$), $v(1-\beta) = s + R$, with $v = B/P$ and $R = \tilde{\nu}\,\lambda\bar{V}$ from (19). Holding P fixed and B predetermined holds v fixed, so the identity requires $ds + dR = 0$. Substituting $dR/d\lambda = \tilde{\nu}\bar{V} > 0$ from Proposition 1 (with $\tilde{\nu}$ fixed by the pricing relation, independent of λ) gives $ds/d\lambda = -\tilde{\nu}\bar{V} < 0$. □

The augmented identity (18) is an equilibrium condition, not a behavioral rule, and it is silent on which variable adjusts to make it hold. The claim that P_t (and, through nominal rigidities, real activity) is the variable that clears the identity, is the subject of the determinacy conditions and alternative model formulations.

To let the price level carry real effects, output is endogenized with a production side and Calvo pricing. The derivation is standard (Galí, 2015) and is given in Appendix 2; it yields the New Keynesian Phillips curve

$$\pi_t = \beta\,\mathbb{E}_t\pi_{t+1} + \kappa\sigma\,\hat{c}_t, \qquad \kappa = \frac{(1-\theta)(1-\beta\theta)}{\theta}, \tag{22}$$

with the resource constraint carried separately. Log-linearizing the short Euler (7) gives the IS curve

$$\hat{c}_t = \mathbb{E}_t\hat{c}_{t+1} - \frac{1}{\sigma}(\hat{\imath}_t - \mathbb{E}_t\hat{\pi}_{t+1})\,, \tag{23}$$

and a positive gross trend $\bar{\pi}$ is accommodated by indexing non-reoptimizing firms to $\bar{\pi}$, so (22) keeps its form with π_t read as the deviation from trend. The remaining rules are

$$\hat{\imath}_t = \phi_\pi\hat{\pi}_t + u_t^i \qquad \text{(monetary policy)}, \tag{24}$$

$$b_t^L = \bar{\chi}(b_t^S + b_t^L) \implies \hat{b}_t^L = \hat{b}_t^S \qquad \text{(maturity)}, \tag{25}$$

$$y_t = c_t + g_t \qquad \text{(resource)}, \tag{26}$$

with u_t^i a policy shock and $\bar{\chi}$ the fixed maturity share.

Definition 4 (Rational-expectations equilibrium)**.** Given parameters, exogenous processes $\{\lambda_t,\, g_t,\, s_t^{\text{exo}},\, u_t^i\}$ and predetermined stocks $\{b_{t-1}^S, b_{t-1}^L\}$, with the primary surplus s_t determined by the fiscal stance specified below (exogenous in Model I, by the feedback rule (46) in Model II) an equilibrium is a collection of bounded processes for the endogenous variables such that (i) the household conditions (7), (8), the binding coverage constraint (5), and slackness (9) hold; (ii) firms' optimal pricing and the price index hold, yielding the Phillips curve (22); (iii) the fundamental price (10), the wedge, and the rent hold; (iv) the flow budget (11), the recursive valuation identity (15), and the monetary, maturity, and resource conditions (24)–(26) hold; and (v) the transversality conditions hold,

$$\lim_{T\to\infty}\mathbb{E}_t\left[\beta^T c_T^{-\sigma} Q_T^S b_T^S\right] = 0, \qquad \lim_{T\to\infty}\mathbb{E}_t\left[\left(\prod_{k=1}^{T} M_{t+k}\right) Q_T^L b_T^L\right] = 0. \tag{27}$$

The two conditions in (27) are the household's no-Ponzi conditions on the short and long bonds. Since $Q_T^L b_T^L$ and $Q_T^S b_T^S$ together comprise $\tilde{v}_T$, and $\tilde{v}_T = v_{T+1}\pi_{T+1} - s_{T+1}$ shifts into v, they imply the consolidated no-bubble condition (16) and hence the augmented identity (17). The transversality conditions are therefore what turn the iterated flow budget into the valuation identity that selects the price level; in the linearized system they are imposed by discarding the explosive eigendirections in the generalized-Schur solution.

Whether a bounded equilibrium exists and is unique depends jointly on the monetary and fiscal stances (Leeper, 1991): determinacy requires exactly one active authority and one passive authority. I present the two alternative formulations.

Model I: With $\phi_\pi < 1$, monetary policy is passive and fiscal policy is active. The primary surplus is exogenous and the FTPL identity selects the equilibrium price level.

Model II: With $\phi_\pi > 1$, monetary policy is active and fiscal policy must be passive. The primary surplus follows a debt-stabilizing rule:

$$\hat{s}_t = \gamma_v \hat{v}_{t-1} + \hat{s}_t^{\text{exo}} \tag{28}$$

so the augmented valuation identity is satisfied through surplus adjustment, the Taylor principle pins inflation, and the rent substitutes for taxes with no price-level consequence.

We observe that the same rent from repression and the same enlarged backing has price level effects depending on the regime.

Proposition 3 (Price-level effect under fiscal dominance)**.** *Suppose the economy is in the passive-money, active-fiscal regime ($\phi_\pi < 1$, surplus exogenous). Then the price level is the variable that clears the augmented identity, and a tightening of the regulatory mandate lowers it:*

$$P = \frac{B(1-\beta)}{s+R}, \qquad \left.\frac{\partial P}{\partial \lambda}\right|_s = -\frac{B(1-\beta)\,\tilde{\nu}\bar{V}}{(s+R)^2} < 0. \tag{29}$$

Proof. Under active fiscal the surplus s is set independently of the debt, and under passive money the Taylor principle fails, so the monetary block does not pin the price level; the price level is therefore the only variable free to clear the steady-state identity $v(1-\beta) = s + R$. With $v = B/P$ as defined above, solving gives the first equality. Differentiating, holding s fixed (exogenous under active fiscal) and $\tilde{\nu}$ fixed by the pricing relation as noted, and using $dR/d\lambda = \tilde{\nu}\bar{V}$ from Proposition 1,

$$\frac{\partial P}{\partial \lambda} = -\frac{B(1-\beta)}{(s+R)^2}\frac{dR}{d\lambda} = -\frac{B(1-\beta)\,\tilde{\nu}\bar{V}}{(s+R)^2} < 0, \tag{30}$$

since every factor is positive in the binding regime. □

Three qualifications fix the standing of the propositions in this section. First, the regime is adopted, not derived: fiscal dominance is the maintained hypothesis, confirmed determinate at the calibration of the next section, so the proofs establish a conditional result, not the regime itself. Second, the comparative statics are partial derivatives of the steady-state identity, holding fixed two objects that move in general equilibrium. They hold $\tilde{\nu}$ fixed, but the feedback of λ on $\tilde{\nu}$ through the wedge is second-order at the calibrated steady state and does not reverse the sign. They also hold the inherited nominal debt $B \equiv B^S_{t-1} + (1 + \delta Q^L_t)B^L_{t-1}$ fixed, which suppresses the long-bond revaluation channel that operates through Q^L_t; this is the standard Cochrane (2001, 2023) reading of the long-debt valuation identity, and it is innocuous in the cross-steady-state comparison, where the wedge ω and the fundamental price $Q^{L,f}$ are calibrated and the mandate λ is absorbed on the quantity margin. The revaluation channel is not absent from the mechanism, but it is a feature of the dynamic transition rather than of the steady-state sign, which is what these propositions establish. Third, the propositions sign the destination of the price level, not the path: under nominal rigidity the same destination is reached gradually, with the flexible-price case appearing there as the $\theta \to 0$ limit. The effect on real variables is carried by the Phillips curve and shown through the impulse responses in the next section.

Model I is the fourteen conditions below in the fourteen endogenous variables

$$\{c,\, y,\, \pi,\, i,\, Q^S,\, Q^L,\, Q^{L,f},\, \omega,\, \tilde{\nu},\, R,\, v,\, b^S,\, b^L,\, \tau\},$$

given exogenous $\{\lambda, g, s, u^i\}$. Model II appends (28), making s endogenous: fifteen equations in fifteen variables, with $\phi_\pi > 1$.

#	condition	reference
1	$Q_t^S = \mathbb{E}_t[M_{t+1}/\pi_{t+1}]$	(7)
2	$1 + i_t = 1/Q_t^S$	(7)
3	$Q_t^L(1 - \tilde{\nu}_t) = \mathbb{E}_t[M_{t+1}(1 + \delta Q_{t+1}^L)/\pi_{t+1}]$	(8)
4	$Q_t^{L,f} = \mathbb{E}_t[M_{t+1}(1 + \delta Q_{t+1}^{L,f})/\pi_{t+1}]$	(10)
5	$1 + \omega_t = Q_t^L/Q_t^{L,f}$	Def. 2
6	$Q_t^L b_t^L = \lambda_t \bar{V}$	(5)
7	$R_t = \tilde{\nu}_t Q_t^L b_t^L$	Def. 3
8	$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \sigma \hat{c}_t$	(22)
9	$\hat{\imath}_t = \phi_\pi \hat{\pi}_t + u_t^i$	(24)
10	$y_t = c_t + g_t$	(26)
11	$b_t^L = \bar{\chi}(b_t^S + b_t^L)$	(25)
12	$v_t = [b_{t-1}^S + (1 + \delta Q_t^L) b_{t-1}^L]/\pi_t$	(11)
13	$v_t = s_t + R_t + \mathbb{E}_t[M_{t+1} v_{t+1}]$	(15)
14	$\tau_t = s_t + g_t$	(11)
Model II only:		
15	$\hat{s}_t = \gamma_v \hat{v}_{t-1} + \hat{s}_t^{\text{exo}} \quad (\phi_\pi > 1)$	(46)

4. Calibration and Solution

To find the deterministic steady state, I set the shocks to zero and let consumption be constant, so $\pi = \bar{\pi}$ and $M = \beta$. The short bond prices at $Q^S = \beta/\bar{\pi}$, a net nominal rate $1 + i = \bar{\pi}/\beta$. The fundamental long price solves $Q^{L,f} = \beta(1 + \delta Q^{L,f})/\bar{\pi}$, i.e.

$$Q^{L,f} = \frac{\beta}{\bar{\pi} - \beta\delta}, \tag{31}$$

and the observed price carries the wedge, $Q^L = (1 + \omega)Q^{L,f}$. Dividing the observed long Euler (8) by (31) and solving for the multiplier,

$$\tilde{\nu} = \frac{\bar{\pi} - \beta\delta}{\bar{\pi}} \cdot \frac{\omega}{1 + \omega}. \tag{32}$$

The factor $(\bar{\pi} - \beta\delta)/\bar{\pi}$ is a perpetuity correction (equal to $1 - \beta\delta$ at $\bar{\pi} = 1$): a long-maturity bond needs only a small per-period multiplier to support a sizable price wedge, which is why $\tilde{\nu}$ is small and its reciprocal $1/\tilde{\nu}$ governs the stiff direction of the dynamics. Define the duration coefficients $\eta \equiv \delta Q^L/(1 + \delta Q^L)$ and $\eta_f \equiv \delta Q^{L,f}/(1 + \delta Q^{L,f}) = \beta\delta/\bar{\pi}$. Normalizing total debt market value to one with long share w_L gives $Q^L b^L = w_L$ and $Q^S b^S = 1 - w_L$; the inherited-debt value is $v = b^S + (1 + \delta Q^L) b^L$, with long value share $w_L^v = (1 + \delta Q^L) b^L/v$ and $w_S^v = 1 - w_L^v$. The rent is $R = \tilde{\nu}\, Q^L b^L$, and the steady-state identity $v(1 - \beta) = s + R$ pins the surplus residually as $s/v = (1 - \beta) - R/v$.

4.1. **Log-linearization around the steady state.** Hats denote log-deviations, except π, i (net rates) and $\omega, \tilde{\nu}$ (small levels), which are level deviations. The coefficients $(\eta, \eta_f, \kappa, w_S^v, w_L^v)$ are the steady-state objects just defined, evaluated at the calibration. Linearizing the equilibrium conditions gives the system below.

$$\text{(IS)} \quad \hat{c}_t = \mathbb{E}_t\hat{c}_{t+1} - \tfrac{1}{\sigma}(\hat{\imath}_t - \mathbb{E}_t\hat{\pi}_{t+1}), \tag{33}$$

$$\text{(Phillips)} \quad \hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \kappa\sigma\hat{c}_t, \tag{34}$$

$$\text{(policy)} \quad \hat{\imath}_t = \phi_\pi\hat{\pi}_t + u^i_t, \tag{35}$$

$$\text{(fund. price)} \quad \hat{Q}^{L,f}_t = -\hat{\imath}_t + \eta_f\,\mathbb{E}_t\hat{Q}^{L,f}_{t+1}, \tag{36}$$

$$\text{(obs. price)} \quad \hat{Q}^L_t = \frac{\hat{\nu}_t}{1-\tilde{\nu}} - \hat{\imath}_t + \eta\,\mathbb{E}_t\hat{Q}^L_{t+1}, \tag{37}$$

$$\text{(wedge)} \quad \hat{\omega}_t = (1+\omega)\big(\hat{Q}^L_t - \hat{Q}^{L,f}_t\big), \tag{38}$$

$$\text{(coverage)} \quad \hat{Q}^L_t + \hat{b}^L_t = \hat{\lambda}_t, \tag{39}$$

$$\text{(rent)} \quad \hat{R}_t = \hat{\nu}_t/\tilde{\nu} + \hat{\lambda}_t, \tag{40}$$

$$\text{(maturity)} \quad \hat{b}^L_t = \hat{b}^S_t, \tag{41}$$

$$\text{(resource)} \quad \hat{y}_t = (1-s_g)\hat{c}_t + s_g\hat{g}_t, \tag{42}$$

The inherited-debt value linearizes to

$$\hat{v}_t = w^v_S(\hat{b}^S_{t-1} - \hat{\pi}_t) + w^v_L(\hat{b}^L_{t-1} + \eta\hat{Q}^L_t - \hat{\pi}_t), \tag{43}$$

in which the coefficient on $\hat{\pi}_t$ is $-(w^v_S + w^v_L) = -1$: a one-percent inflation surprise erodes real debt one-for-one, and the long value share w^v_L is the weight on long-bond revaluation, the channel through which a higher Q^L raises the backing. Finally the forward valuation (15) linearizes, after substituting $\hat{M}_{t+1} = -(\hat{\imath}_t - \mathbb{E}_t\hat{\pi}_{t+1})$ and leading (43), to

$$\hat{v}_t = \frac{s}{v}\hat{s}_t + \frac{R}{v}\hat{R}_t + \beta\big(w^v_S\hat{b}^S_t + w^v_L\hat{b}^L_t + w^v_L\eta\,\mathbb{E}_t\hat{Q}^L_{t+1}\big) - \beta\hat{\imath}_t, \tag{44}$$

with $\hat{v}_t$ a state, not a jump (its forward term cancels through (43)).

The primary surplus $s = \tau - g$ is treated as the consolidated fiscal primitive. In Model I it follows an exogenous AR(1),

$$\hat{s}_t = \rho_s\,\hat{s}_{t-1} + \varepsilon^s_t, \tag{45}$$

while in Model II it obeys the debt-stabilizing rule

$$\hat{s}_t = \gamma_v\,\hat{v}_{t-1} + \hat{s}^{\text{exo}}_t, \qquad \gamma_v > 0 \text{ large enough to stabilize debt}, \tag{46}$$

which makes fiscal policy passive. Government spending enters the equilibrium only through the resource constraint (and its own shock $\hat{g}_t$); the tax and expenditure components of the surplus are not propagated separately. Because the surplus reaches the valuation identity (44) only through the backing term $(s/v)\,\hat{s}_t$, the calibrated feedback coefficient $\gamma_v = 10$ is the surplus's response to lagged debt, large only because the steady-state share s/v is small.

This linear system, solved by the generalized-Schur method, is the object the calibration below fills.

This section assigns numerical values, solves for the deterministic steady state, and reports the solved equilibrium as a linear policy function. Parameters are grouped by the source of their discipline; the steady state is an output of those choices, not an object assigned. Determinacy of the system is confirmed numerically at the baseline.

4.2. **Parameters.** Four parameters are deep preferences and technology, fixed at conventional quarterly values. The discount factor is $\beta = 0.99$ (a real rate near one percent per quarter); utility is logarithmic, $\sigma = 1$; the Calvo non-reset probability is $\theta = 0.75$, an average price duration of one year, standard in the New Keynesian literature (Galí, 2015, ch. 3); and the inverse Frisch elasticity is set to $\varphi = 0$, a modeling choice that shuts the hours margin and reduces real marginal cost to $\sigma\hat{c}_t$. The long bond's geometric-perpetuity form follows Woodford (2001); its decay δ is not a free convention but is disciplined by the observed gilt duration. The variety elasticity ε enters only the steady-state markup and drops out of the linearized dynamics, so it is left unassigned.

symbol	value	meaning	basis
β	0.99	discount factor	$\approx$ 1% quarterly real rate
σ	1	inverse IES / CRRA	log utility
θ	0.75	Calvo non-reset probability	one-year price duration (Galí, 2015)
φ	0	inverse Frisch elasticity	no hours margin (choice)

Three parameters, and the value of δ, are mapped to UK observables. The decay $\delta = 0.998$ gives the perpetuity a Macaulay duration $D = 1/(1 - \beta\delta/\bar{\pi}) \approx 14.8$ years, matching the long-end gilts held by the captive pension-and-insurance sector under liability-driven investment (Bloomberg UK Government 15+ Year Index, reported in the Vanguard U.K. Long Duration Gilt Index Fund fact-sheet, April 2026: duration 15.1 years, maturity 25.9 years). The long share of debt by market value is $w_L = 0.353$, the fraction of marketable gilts in the 15+ year bucket (UK Debt Management Office, gilts in issue by maturity band at market value, 2024–25). The government-purchases share is $s_g = 0.40$ (Office for National Statistics and Office for Budget Responsibility, 2024–25, current expenditure $\approx$ 40% of GDP); because the Phillips curve is written in consumption, s_g enters only the resource constraint and the equilibrium is invariant to it.

symbol	value	meaning	mapping
δ	0.998	long-bond geometric decay	$D \approx 14.8$ yr $\leftrightarrow$ 15+ gilt index
w_L	0.353	long share of debt (market value)	DMO maturity split (2024–25)
s_g	0.40	government purchases / output	ONS/OBR current expenditure (2024–25)

The remaining choices fix the regime and two scales. The baseline is passive-money / active-fiscal: $\phi_\pi = 0.5 < 1$, so the price level is pinned by the government budget rather than the monetary rule. Trend inflation is the Bank of England's two-percent annual target, $\bar{\pi} = (1.02)^{1/4} = 1.005$ in quarterly terms. Regulatory intensity has no time series; regulatory regimes are slow-moving, so $\rho_\lambda = 0.95$ is set, corresponding to a 3.5 half life of a regulatory change. The other persistences are estimated posterior means (Smets and Wouters, 2007). The captive liability pool is normalized to $\bar{V} = 1$, and the fiscal-feedback coefficient $\gamma_v = 10$ is used only in the active-money variant. The surplus shock persistence is set to 0.9, within the range estimated by Bohn (1998). Because the surplus process and the regulatory shock are orthogonal exogenous drivers, the propagation of surplus innovations alone and does not enter the regulatory-shock impulse responses or the steady-state frontier that constitute the paper's results; it is reported for completeness.

symbol	value	meaning	basis / role
ϕ_π	0.5	Taylor coefficient	< 1: passive money (baseline)
$\bar{\pi}$	1.005	gross quarterly trend inflation	BoE 2%/yr: $(1.02)^{1/4}$
ρ_λ	0.95	regulatory-shock persistence	set; swept in robustness
ρ_g	0.97	spending-shock persistence	Smets–Wouters (2007)
ρ_s	0.9	surplus-shock persistence	Bohn (1998)
ρ_i	0.15	monetary-shock persistence	Smets–Wouters (2007)
$\bar{V}$	1	captive liability pool	normalization
γ_v	10	fiscal-feedback coefficient	active-money variant only

The parameter ω deserves deeper discussion. It represents the difference between the actual, observed price of the captive bond and the counterfactual price it would have unconstrained; its "fundamental" price.

The fundamental price is inherently unobservable so any empirical counterpart must be constructed. A large literature does this for sovereign debt: the "valuation puzzle" of Jiang, Lustig, Van Nieuwerburgh, and Xiaolan (2024) shows the market value of government debt exceeds the present value of its backing surpluses by a margin frictionless models cannot reconcile, and points to a convenience yield to close the gap; Krishnamurthy and Vissing-Jorgensen (2012) measure that premium directly. The wedge ω is the same object, price above frictionless value or yield below frictionless yield, but narrower in scope. Where this literature measures the total premium on sovereign debt as an asset class (bundling safety, liquidity, duration "conveniences"), ω maps to the specific distortion of duration-matching mandates on long-maturity government bonds.

To assign a value to the wedge, we must first establish whether the wedge exists at all. In the model, the existence of a wedge hinges on our set of assumptions. This is, that markets are segmented, that nobody can take an offsetting unconstrained position to arbitrage away the wedge, and that the constraint itself is binding. All in, the price gets determined by the constrained optimality condition.

The segmentation we impose by assumption has an observable counterpart in who holds the long end of the gilt market. Insurance companies and pension funds are dominant holders of UK government debt, on the order of two-thirds of the total stock in the early 2000s (Bank of England, 2002) before the Bank of England started purchasing these bonds under QE and reducing their share. Demand is concentrated on the long-duration maturities they use to match liabilities (ONS; OBR, 2025). Ownership shares are not public split by maturity over time, but private defined-benefit pension fund holdings alone are on the order of a third of the 25y+ market (author's calculation from ONS, 2022, and DMO, 2022), so total captive ownership of the long end (which should add insurers and public pension funds) is higher still.

Being the dominant holders, however does not imply that they are the marginal pricers. A zoom into the long bond market during the crisis of September 2022 gives credence to this claim.

The timeline of September 2022 went as follows: On the 23rd, the budget was announced. Long yields rose, standard FTPL prediction. This rise in yields (drop in market value of the bond portfolio) triggered margin calls on their leveraged positions, forcing them to sell to raise collateral in a self-reinforcing spiral that was only stopped by an emergency intervention by the Bank of England on the 28th. The dynamic between the 23rd and the 28th was entirely a real-yield phenomenon (the up and down was not a repricing of expected inflation) and nobody was stepping in to arbitrage the mispricing away. There had been no changes in fundamentals, which only happened a bit later, in the first week of October, when we can see that yields renormalized to the before-budget level. The initial rise on the 23rd was a genuine response to the fiscal news; the non-fundamental claim rests on the spiral between the 23rd and 28th, and the reversal against an unchanged fiscal stance.

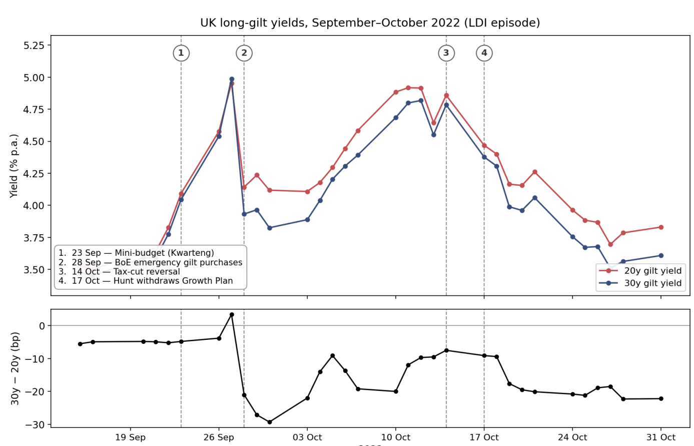

FIGURE 1. Timeline of the long sovereign bond market during the September 2022 crisis

Interestingly, this was a Quantitative Easing-like intervention (monetary authority stepping into directly purchase government bonds in the secondary market) but during a period of contractive Quantitative Tightening. Through the lens of this model, an operation justified on financial stability grounds also has a price-level interpretation: arresting the forced unwind prevents the wedge (and with it the rent backing the debt) from collapsing.

I do not claim to identify the size of the wedge with this exercise. My only claim is that this episode shows that the mechanism that generates it is real. Price is, on the margin, determined by the specific dynamic within the class of captive bondholders (i.e the pension funds).

There is also evidence from the 2004-05 pension reform, when the liability-matching regulation was tightened. Greenwood and Vayanos (2010) studied this mechanism specifically, showing in an event study of the 2004 reform that the forced, duration-matching demand from pension funds compressed long-term gilt yields relative to short, most pronounced at the longest maturities, establishing the sign of the effect and its regulatory origin, though not a clean per-period magnitude.

I propose to use the difference between the 30y and the 20y yield during the pre-financial-crisis period as a measure of the regulatory wedge. In this period the yield curve was inverted at the long end. There are three standard explanations for such an inversion: expectations of falling short rates, convexity, and segmented markets under preferred habitat. The first requires the expected average short rate over years 20–30 to lie below that over the nearer horizon, and there is no economic basis for a downward-sloping expected-short-rate path that far out; absent such a story, expectations cannot deliver a persistent, signed gap of this size. The second, convexity, scales with the level of rate volatility, yet the deep inversion of 2006–07 is relatively low-volatility compared to the periods that follow which display no comparable inversion: the spread does not track volatility as a convexity effect would predict. This leaves segmented markets and preferred habitat, the argument I lean on. The residual gap conflates the wedge with a term premium, but since the term premium is signed the opposite way—pushing the 30y above the 20y—it works against the inversion, so the measured spread understates the wedge and can be read as a floor, under an assumption of a zero term premium. This is why I cannot rely on the post-financial crisis period where the yield curve regained a positive slope. See Appendix 3 for more detail.

The average 30y to 20y yield spread between 2006 and 2007 was around 23 bp.

$$\omega \approx D \times \mathit{spread} = 14.8 \times 0.0023 \approx 0.034 \tag{47}$$

This mapping is far from perfect. The distortion does not occur on the 30y specifically, the model simplifies the long portion into a Woodford perpetuity, and we established that the measurement is only a floor. As such, the claim is only that the calibration is set to study a directionally illustrative effect and not by any means a proper quantification of the effect of changes in regulation on inflation, nor of the fiscal space gained by financial repression.

4.3. **Steady state.** Given the parameters and the recovered wedge, the deterministic steady state is solved. The short bond prices at $Q^S = \beta/\bar{\pi} = 0.985$; the fundamental and observed long prices are $Q^{L,f} = 58.30$ and $Q^L = (1+\omega)Q^{L,f} = 60.29$; the multiplier is $\tilde{\nu} = 0.00056$, small because the perpetuity-correction factor spreads the wedge over the bond's long life (its reciprocal $1/\tilde{\nu} \approx 1800$ is the stiff direction in the dynamic system).

object	expression	value
short price	$Q^S = \beta/\bar{\pi}$	0.985
net nominal rate	$i = \bar{\pi}/\beta - 1$	1.52%
fundamental long price	$Q^{L,f} = \beta/(\bar{\pi} - \beta\delta)$	58.30
observed long price	$Q^L = (1+\omega)Q^{L,f}$	60.29
multiplier	$\tilde{\nu}$	0.00056
Phillips slope	$\kappa = (1-\theta)(1-\beta\theta)/\theta$	0.086
value shares	(w_S^v, w_L^v)	(0.647, 0.353)
rent share	R/v	0.019%
primary-balance share	$s/v = (1-\beta) - R/v$	0.981%
duration	$D = 1/(1-\beta\delta/\bar{\pi})$	14.8 yr

The calibration satisfies the steady-state backing identity exactly, $s/v + R/v = 0.981\% + 0.019\% = 1.00\% = 1 - \beta$: the per-period flow servicing the debt splits between the primary balance and the repression rent, and the two sum to the net discount. This is the calibration's internal unit test.

The log-linearized equilibrium is cast in the canonical form of Sims (2002)

$$G_0 y_t = G_1 y_{t-1} + \Psi \varepsilon_t + \Pi \eta_t \tag{48}$$

with y_t stacking the fourteen endogenous variables, the four exogenous AR(1) drivers, and four expectation terms; forward expectations are replaced by auxiliary variables tied to their realizations by forecast-error identities.

The system is solved by the generalized-Schur (QZ) decomposition. A unique bounded solution exists when the number of explosive roots equals the number of forecast errors (four). . At the baseline this holds: the solver returns existence/uniqueness flags `eu` = $[1, 1]$. (The row-by-row mapping of equations to (G_0, G_1) and the QZ ordering convention are in Appendix 4)

Solving yields the linear policy function

$$y_t = \Theta_1 y_{t-1} + \Theta_0 \varepsilon_t \tag{49}$$

which expresses every variable as a function of a five-dimensional minimal state: one predetermined endogenous object (inherited debt, the two bond stocks collapsing to one through the maturity identity $\hat{b}_t^L = \hat{b}_t^S$) and the four exogenous drivers. The price level, consumption, both long-bond prices, and the rent are all jumps.

5. Results

The calibrated model is now put to work on the two questions that motivate it. The first is positive: what happens to inflation, the price level, and the real economy when the regulatory mandate tightens. The second is the constrained-policy question that gives the mechanism its fiscal meaning: how far the primary surplus can fall as repression intensifies.

Both are readings of the single policy function solved above. Reading its regulatory-driver rows for a ten-percent tightening of the mandate gives:

variable	impact	long-run
price level $\hat{P}$	-0.097%	-0.589%
long-bond price $\hat{Q}^L$	$+9.96\%$	—
real debt value $\hat{v}$	$+3.55\%$	—
rent contribution $(R/v)\hat{R}$	$+0.22\%$ of debt	—
consumption $\hat{c}$	-0.197%	—

The tightening raises the long-bond price almost one-for-one (captive demand is price-inelastic), revalues the debt upward, and lowers the price level: the rent substitutes for inflation as fiscal backing. Everything that follows is this same policy function read in a different direction. The impulse response iterates it forward from a single innovation; the fiscal-space frontier takes the present value of the rent row under a permanent change. The section closes by showing the two are the same object viewed from two sides.

The effect is notably small, by construction. It prices a single repression instrument: duration-matching mandates on long-maturity bonds, and only that fraction of the maturity structure they bind on (roughly a third of the stock). It is calibrated, moreover, to a deliberately conservative 23 bp wedge, established as a floor rather than a point estimate. What the exercise isolates is therefore one cell of a much larger taxonomy: duration matching is a single lever among the safety, liquidity, capital, and collateral-eligibility distortions that Reis (2025) catalogues across the rest of the sovereign balance sheet, none of which the model carries. The consolidated repression rents estimated empirically are correspondingly larger by orders of magnitude. The results obtained here are not a failure to reproduce that figure but the expected size of one narrow channel priced in isolation: the marginal contribution of a single instrument on a single tranche, at a floor wedge. Read this way, the per-period rent is a lower bound on the fiscal pull of repression, and the channel's economic interest lies in its existence and its regime-dependence rather than in the magnitude of this one calibrated slice.

5.1. **Impulse response to a regulatory shock.** Figure 2 traces the response to a ten-percent tightening of the mandate ($\hat{\lambda}_0 = 0.10$, decaying at $\rho_\lambda = 0.95$) under the passive-money baseline. The mandate forces the captive sector to hold a larger value of long bonds; because its demand is price-inelastic, the adjustment falls on the price, not the quantity. The long-bond price rises $\hat{Q}_0^L = +9.96\%$, the constraint binds harder ($\hat{\tilde{\nu}}_0 = +0.64\%$), and the wedge widens $\hat{\omega}_0 = +10.0\%$. A constraint that binds harder extracts more rent: the per-period rent's raw log-deviation is large ($\hat{R}_0 \approx +1160\%$) only because its steady-state base is tiny, so the meaningful object is its contribution to backing, $(R/v)\hat{R}_0 = +0.22\%$ of debt on impact. This rising rent is Proposition 1 in the impulse response: the binding mandate relaxes the budget constraint, $\partial R/\partial\lambda = \tilde{\nu}\bar{V} > 0$.

That extra backing must be absorbed. With the surplus exogenous in this regime, the augmented identity splits it between a higher real debt value and a lower price level: the real value of debt rises $\hat{v}_0 = +3.55\%$, inflation falls $\hat{\pi}_0 = -0.097\%$, and the price level settles permanently lower, by 0.589%. Almost all of that impact rise is revaluation, not disinflation: decomposing (44) with the lagged stocks predetermined gives $\hat{v}_0 = w_L^v\, \eta\, \hat{Q}_0^L - \hat{\pi}_0 = +3.46\% + 0.10\%$, so the upward revaluation of the inherited long bonds is 97% of the move. The revaluation channel that the steady-state propositions hold fixed (the numerator B of (18)) is thus the dominant driver of debt-value dynamics on the transition: the propositions sign the destination, and this shows the path is carried by revaluation. The adjustment is gradual, about seventeen percent of it on impact, the rest unfolding over several years as sticky prices reset. The nominal rate falls passively ($\hat{\imath}_0 = -0.049\%$, since $\phi_\pi < 1$), and the slow disinflation drags the real economy down, $\hat{c}_0 = -0.197\%$ and $\hat{y}_0 = -0.118\%$.

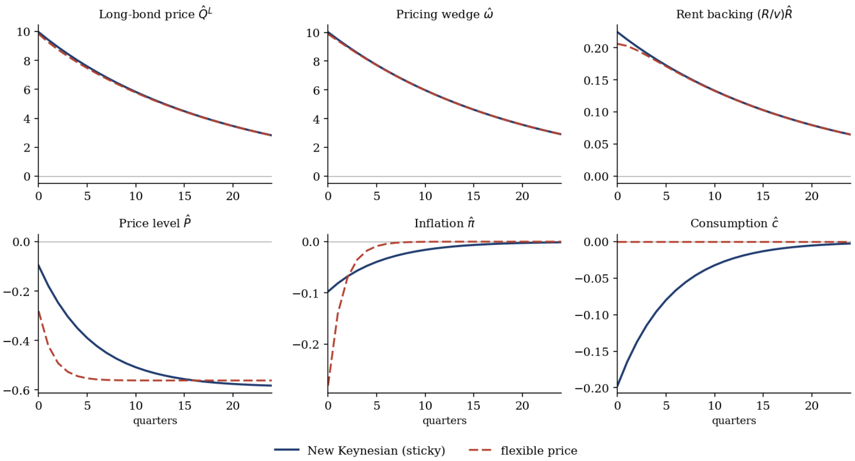

FIGURE 2. Impulse responses to a ten-percent tightening of the regulatory mandate ($\rho_\lambda = 0.95$) under passive money. Solid: New Keynesian (sticky prices); dashed: the flexible-price benchmark. The rent is shown as its contribution to backing $(R/v)\hat{R}$, not its raw log-deviation.

The sign pattern is the result, and it is Proposition 3 as a dynamic path rather than a steady-state derivative: a tighter mandate lowers the price level, the rent doing the work that surprise inflation would otherwise do. The flexible-price benchmark isolates the role of rigidity. There, the price level adjusts almost immediately and the real allocation is untouched ($\hat{c}_0 \approx 0$): the economy simply reprices. Sticky prices convert that repricing into a drawn-out disinflation with real costs.

5.2. **Fiscal space.** The impulse response holds the surplus fixed and lets the price level move. The dual exercise holds inflation fixed and asks how far the surplus can fall as repression intensifies. In steady state the flow servicing the debt splits between the primary balance and the rent,

$$\frac{s}{v} = (1-\beta) - \frac{R}{v}, \tag{50}$$

and the two sum to the net discount $1-\beta$. Every unit of rent the mandate generates is a unit by which the required surplus may fall: (50) is Proposition 2 traced out, the comparative static $ds/d\lambda|_P = -\tilde{\nu}\bar{V}$ read across steady states.

Panel (a) of Figure 3 plots it. At the calibration the economy sits near the top-left corner: $R/v = 0.019\%$, so repression covers about two percent of the debt-service flow while the surplus carries $s/v = 0.981\%$. The balanced-budget threshold lies at $R/v = 1-\beta = 1\%$, an order of magnitude beyond what the United Kingdom runs. Read per period, repression of the observed magnitude is a marginal instrument: it trims the surplus at the edge, no more. Panel (b) capitalizes the rent over the bond's fifteen-year life: a ten-percent permanent tightening creates fiscal space worth 0.19% of debt, a twenty-five-percent tightening 0.48%, a fifty-percent tightening 0.97%.

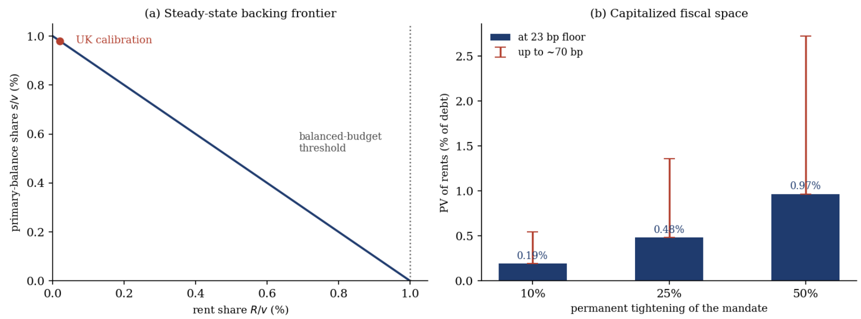

FIGURE 3. Fiscal space. Panel (a): the steady-state frontier $s/v = (1 - \beta) - R/v$, with the UK calibration (red) and the balanced-budget threshold. Panel (b): the present value of the rents from a permanent tightening, at the 23-bp wedge floor (bars) with the band up to ~ 70 bp (whiskers).

The two exercises are one object read twice. The augmented identity fixes the inherited real debt and backs it with surpluses and rents; when the mandate manufactures rent, that backing is absorbed somewhere. Hold the surplus fixed and the price level falls; hold inflation fixed and the surplus falls. Deflation and a freed surplus are not two mechanisms but the same repression backing on two margins: the choice between them is a choice of nominal anchor, not a difference in the resource transfer.

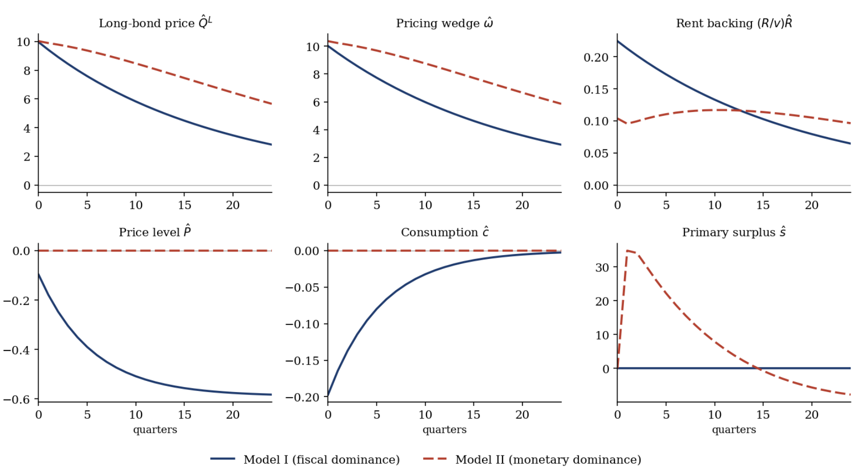

FIGURE 4. Regime contrast. Responses to a ten-percent tightening under fiscal dominance (Model I, solid) and monetary dominance (Model II, dashed). The asset-pricing block (top) is regime-invariant; the destination of the backing (bottom) is not. Shown over twenty-four quarters; stationary variables revert to trend, while the fiscal-dominance price level settles at a lower level.

5.3. **Active money.** Whether the backing transmits to prices is a property of the regime, not the mechanism. Figure 4 sends the identical tightening through the active-money, passive-fiscal configuration (Model II: $\phi_\pi = 1.5$, surplus following (28), $\gamma_v = 10$). The asset-pricing block is regime-invariant: the long-bond price still jumps $\hat{Q}_0^L = +10.0\%$, the wedge widens, and the constraint extracts rent of comparable order ($(R/v)\hat{R}_0 = +0.10\%$ of debt). With the Taylor principle pinning the price level, the surplus is the passive margin: the debt-stabilizing rule raises it sharply ($\hat{s}_1 \approx +35\%$ off its small base) to service the revaluation-swollen debt, then reverts, while the price level and real allocation are essentially undisturbed ($\hat{\pi}_0 \approx 0$, $\hat{c} \approx 0$). The same rent and the same backing produce a disinflation under fiscal dominance and a tax cut under monetary dominance: a placebo that attributes the price effect to the regime, not to repression as such.

6. Conclusion

This dissertation gives a structural microfoundation to convenience yields as financial repression (or, the shadow price of a binding duration-matching mandate), embedded inside a fiscal-dominance model with price rigidities. The rent the sovereign gets from the wedge in bond prices they generate through regulation enters the valuation identity as genuine backing alongside the primary surplus. The channel it produces is conditional but straightforward: regulation relaxes the budget constraint as an accounting property (Proposition 1), and whether that backing reaches the price level turns on equilibrium selection: under fiscal dominance a tighter mandate lowers the price level (Proposition 3), with the real economy bearing the adjustment through sticky prices, while under monetary dominance the surplus absorbs the rent and prices are unmoved (Proposition 2). The model is then calibrated to UK data, and the exercise remains directional and illustrative. No claims to identification or quantification of the actual size of the proposed effect are made.

The model's reach is bounded by its reduced form. A single stochastic discount factor prices both bonds, so the wedge is a constrained optimality condition rather than the outcome of genuinely segmented balance sheets; and because the captive arm belongs to the representative household, the distributional incidence, which is the heart of the political-economy story, remains unmodeled. Moreover, the mandate is exogenous, and the monetary-fiscal regime is imposed to condition the results. The natural extensions address each of these shortcomings in turn: a two-kernel segmentation that prices the long bond off a distinct, balance-sheet-constrained intermediary; an endogenous mandate chosen by a sovereign trading off repression against taxation, in the spirit of the optimal-repression literature; and alternative avenues for calibration.

What the dissertation establishes is the coherence of the channel: prudential regulation, justified on grounds orthogonal to fiscal policy, can move inflation and output through the backing of the public debt.

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7. Appendix 1: Household Problem

7.1. **Household first-order conditions.** Attach the multiplier $\beta^t\Lambda_t$ to the flow budget (4) and $\beta^t\nu_t$, $\nu_t \geq 0$, to the coverage constraint (5) written as $Q_t^L b_t^L - m_t \geq 0$:

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Bigg[\frac{c_t^{1-\sigma}}{1-\sigma} - \psi\,\frac{N_t^{1+\varphi}}{1+\varphi} - \Lambda_t\Big(c_t + Q_t^S b_t^S + Q_t^L b_t^L - y_t + \tau_t - \frac{b_{t-1}^S}{\pi_t} - \frac{(1+\delta Q_t^L) b_{t-1}^L}{\pi_t}\Big) + \nu_t\big(Q_t^L b_t^L - m_t\big)\Bigg]. \tag{51}$$

The choice variables are $\{c_t, b_t^S, b_t^L\}$. We differentiate term by term, taking care that a date-t choice of a bond appears in the date-t budget (purchase) and the date-$(t+1)$ budget (payoff), the latter carried with the extra discount factor β.

Consumption FOC.. The only terms in (51) containing c_t are the felicity $\beta^t c_t^{1-\sigma}/(1-\sigma)$ and the budget term $-\beta^t\Lambda_t c_t$. Hence

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t\big(c_t^{-\sigma} - \Lambda_t\big) = 0 \quad \Longrightarrow \quad \Lambda_t = c_t^{-\sigma}. \tag{52}$$

The budget multiplier equals the marginal utility of consumption.

Short bond FOC.. The choice b_t^S enters (51) in two places: the date-t budget term $-\beta^t\Lambda_t Q_t^S b_t^S$ and the date-$(t+1)$ budget term $+\beta^{t+1}\Lambda_{t+1} b_t^S/\pi_{t+1}$ (the latter from the maturing-bond term b_{t-1}^S/π_t shifted one period forward, carried with β^{t+1} and under $\mathbb{E}_t$). Setting the derivative to zero,

$$\frac{\partial \mathcal{L}}{\partial b_t^S} = -\beta^t\Lambda_t Q_t^S + \beta^{t+1}\mathbb{E}_t\left[\Lambda_{t+1}\frac{1}{\pi_{t+1}}\right] = 0. \tag{53}$$

Divide through by $\beta^t\Lambda_t$:

$$Q_t^S = \beta\,\mathbb{E}_t\left[\frac{\Lambda_{t+1}}{\Lambda_t}\frac{1}{\pi_{t+1}}\right]. \tag{54}$$

Defining the stochastic discount factor $M_{t+1} \equiv \beta\,\Lambda_{t+1}/\Lambda_t = \beta(c_{t+1}/c_t)^{-\sigma}$ via (52) gives the short-bond Euler equation (7) of the main text.

Long bond FOC.. The choice b_t^L enters (51) in three places: the date-t budget term $-\beta^t\Lambda_t Q_t^L b_t^L$; the date-$(t+1)$ budget term $+\beta^{t+1}\Lambda_{t+1}(1+\delta Q_{t+1}^L)b_t^L/\pi_{t+1}$ (from the maturing long-bond term shifted forward, since one unit held into $t+1$ pays the coupon-plus-continuation value $1+\delta Q_{t+1}^L$); and the coverage term $+\beta^t\nu_t Q_t^L b_t^L$. Setting the derivative to zero,

$$\frac{\partial\mathcal{L}}{\partial b_t^L} = -\beta^t\Lambda_t Q_t^L + \beta^t\nu_t Q_t^L + \beta^{t+1}\mathbb{E}_t\left[\Lambda_{t+1}\frac{1+\delta Q_{t+1}^L}{\pi_{t+1}}\right] = 0. \tag{55}$$

Divide by $\beta^t\Lambda_t$ and define the normalized multiplier $\tilde{\nu}_t \equiv \nu_t/\Lambda_t \geq 0$:

$$Q_t^L - \tilde{\nu}_t Q_t^L = \beta\,\mathbb{E}_t\left[\frac{\Lambda_{t+1}}{\Lambda_t}\frac{1+\delta Q_{t+1}^L}{\pi_{t+1}}\right], \tag{56}$$

which is the long-bond Euler equation (8) of the main text.

8. Appendix 2: NK details

The block is the standard Calvo derivation (Galí, 2015); I record only the steps that fix notation and the two points specific to this model: the $\varphi = 0$ marginal cost and the consumption- rather than output-based Phillips curve.

A competitive final-good producer bundles varieties through $y_t = \left(\int_0^1 y_t(i)^{(\varepsilon-1)/\varepsilon}di\right)^{\varepsilon/(\varepsilon-1)}$, $\varepsilon > 1$. Expenditure minimization gives the usual isoelastic demand and, substituting it back into the aggregator, the price index:

$$y_t(i) = \left(\frac{P_t(i)}{P_t}\right)^{-\varepsilon} y_t, \qquad P_t = \left(\int_0^1 P_t(i)^{1-\varepsilon}di\right)^{\frac{1}{1-\varepsilon}}, \tag{57}$$

where the multiplier on the aggregator constraint is the marginal cost of one unit of the bundle, i.e. the price index P_t.

Firms produce with labor under constant returns, $y_t(i) = A_t n_t(i)$, so at nominal wage W_t nominal marginal cost is common across firms, $MC_t = W_t/A_t$. The household's labor-supply condition $W_t/P_t = \psi N_t^{\varphi} c_t^{\sigma}$ then gives real marginal cost $mc_t \equiv MC_t/P_t = \psi N_t^{\varphi} c_t^{\sigma}/A_t$.

Assumption 4 (No hours margin)**.** $\varphi = 0$.

There remains a labor market and a genuine cost; the assumption only removes the dependence of the real wage on hours. With $\varphi = 0$ and a unit-productivity normalization $A_t = 1$, $mc_t = \psi c_t^{\sigma}$, so

$$\widehat{mc}_t = \sigma\,\hat{c}_t. \tag{58}$$

Real marginal cost tracks consumption: by separability the wage is governed by the marginal utility of consumption and by nothing on the hours side.

Each period a firm re-optimizes with probability $1-\theta$; resetters are a representative sample and, by symmetry, choose a common P_t^*. Discounting a date-$(t+k)$ nominal payoff by $\Lambda_{t,t+k} = \beta^k(c_{t+k}/c_t)^{-\sigma}P_t/P_{t+k}$ and survival by θ^k, the reset price maximizes

$$\mathbb{E}_t\sum_{k=0}^{\infty}\theta^k\Lambda_{t,t+k}\left(P_t^* Y_{t+k|t} - TC_{t+k}(Y_{t+k|t})\right), \qquad Y_{t+k|t} = (P_t^*)^{-\varepsilon}P_{t+k}^{\varepsilon}Y_{t+k}. \tag{59}$$

The demand elasticity in P_t^* is constant, $\partial Y_{t+k|t}/\partial P_t^* = -\varepsilon\,Y_{t+k|t}/P_t^*$; using it and scaling the first-order condition by the positive constant $P_t^*/(\varepsilon-1)$ gives, with gross markup $\mathcal{M} \equiv \varepsilon/(\varepsilon-1)$,

$$\mathbb{E}_t\sum_{k=0}^{\infty}\theta^k\Lambda_{t,t+k}Y_{t+k|t}(P_t^* - \mathcal{M}MC_{t+k}) = 0. \tag{60}$$

At a zero-inflation steady state $Y_{t+k|t} = Y$ and $\Lambda_{t,t+k} = \beta^k$, so the bracket in (60) must vanish:

$$P^* = \mathcal{M}MC \quad \Longrightarrow \quad \mathcal{S} \equiv \frac{MC}{P} = \frac{1}{\mathcal{M}} = \frac{\varepsilon - 1}{\varepsilon}, \qquad \mathcal{M}\mathcal{S} = 1. \tag{61}$$

Let $\widehat{mc}_t \equiv \ln(\mathcal{M}\mathcal{S}_t)$ (zero in steady state, by (61)), $\hat{p}^*_t \equiv \ln(P^*_t/P_{t-1})$, and $\pi_t \equiv \ln(P_t/P_{t-1})$. Linearizing (60) about the zero-inflation steady state: because the bracket vanishes there, all variation in the weight $\theta^k \Lambda_{t,t+k} Y_{t+k|t}$ multiplies zero and drops out, so only the bracket need be linearized. Using $\mathcal{M}MC_{t+k}/P_{t-1} = e^{\widehat{mc}_{t+k} + \sum_{j=0}^{k} \pi_{t+j}}$ and $e^x \approx 1 + x$,

$$\hat{p}^*_t = (1 - \beta\theta)\, \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \left[\widehat{mc}_{t+k} + \sum_{j=0}^{k} \pi_{t+j} \right]. \tag{62}$$

Swapping the double sum (each π_{t+j} enters every term with $k \geq j$) and collapsing the resulting present values to a recursion gives

$$\hat{p}^*_t = (1 - \beta\theta)\, \widehat{mc}_t + \pi_t + \beta\theta\, \mathbb{E}_t \hat{p}^*_{t+1}. \tag{63}$$

The price index (57) splits into Calvo groups, $P_t^{1-\varepsilon} = \theta P_{t-1}^{1-\varepsilon} + (1-\theta)(P^*_t)^{1-\varepsilon}$, which linearizes to

$$\pi_t = (1 - \theta)\hat{p}^*_t. \tag{64}$$

Substituting (64) at t and led one period into (63) and collecting the π_t terms (using $\pi_t - (1-\theta)\pi_t = \theta\pi_t$) gives the New Keynesian Phillips curve

$$\pi_t = \beta\, \mathbb{E}_t \pi_{t+1} + \kappa\, \widehat{mc}_t, \qquad \kappa = \frac{(1-\theta)(1-\beta\theta)}{\theta}. \tag{65}$$

Substituting marginal cost in consumption (58), $\widehat{mc}_t = \sigma \hat{c}_t$,

$$\pi_t = \beta\, \mathbb{E}_t \pi_{t+1} + \kappa\sigma\, \hat{c}_t, \tag{66}$$

with the resource constraint carried separately. (Writing the curve in output, $\widehat{mc} = \sigma\hat{y}$, would misscale the slope by $1/(1 - s_g)$, $s_g \equiv g/y$, and omit a government-spending forcing term.)

9. Appendix 3: Details on Wedge

Figure 5. 30y vs 20y bond yield spread

TABLE 1. Long-end gilt spread and rate volatility, annual means, 2005–2024.

Year	Spread (bp)	Vol (bp)	Year	Spread (bp)	Vol (bp)
2005	−5.1	49.6	2015	16.2	75.2
2006	**−21.0**	**51.5**	2016	14.3	73.9
2007	**−24.5**	**46.0**	2017	6.5	56.1
2008	−26.0	74.5	2018	2.4	51.0
2009	1.6	101.8	2019	9.6	62.1
2010	6.0	67.6	2020	7.2	76.2
2011	13.2	72.0	2021	0.6	64.4
2012	33.0	73.4	2022	−7.8	158.9
2013	30.9	57.4	2023	−0.2	106.8
2014	15.5	47.3	2024	5.6	81.0

Spread is the 30y minus 20y zero-coupon gilt yield; volatility is the annualized standard deviation of daily 30y-yield changes (63-day rolling window). Convexity is worth more the higher the rate volatility, and the 30y carries more convexity than the 20y, so a convexity-driven inversion should be deepest when volatility is highest. The data show the reverse: the deep 2006–07 inversion (−23 bp) sits in the lowest-volatility regime of the sample (≈ 47–51 bp), while the high-volatility years (2008–09, 2022–23) carry near-zero, positive, or far shallower spreads. The cross-year correlation between volatility and the spread is effectively zero, against the strong negative value convexity requires, so the gap reflects segmented-market, preferred-habitat demand rather than convexity.

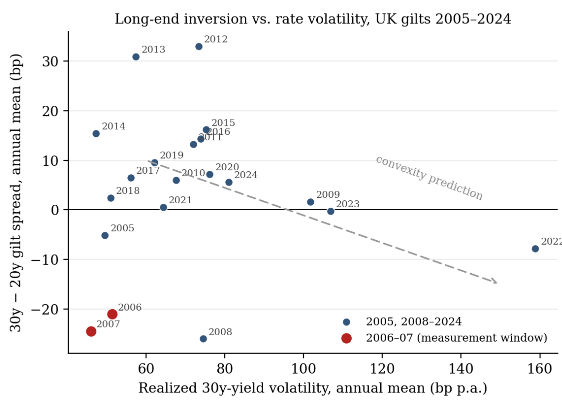

FIGURE 6. Long-end inversion against rate volatility, UK gilts 2005–2024 (annual means). The 2006–07 measurement window is highlighted; the dashed arrow marks the direction convexity predicts. The inversion sits at the low-volatility extreme, opposite to that prediction.

10. APPENDIX 4: COMPUTATIONAL SOLUTION

The model is solved numerically with the Python implementation listed below. It builds the fourteen log-linearized equilibrium conditions, the four exogenous AR(1) drivers, and four expectation terms into a single stacked vector y_t of length $N = 22$, written in the canonical form of Sims (2002),

$$G_0\, y_t \;=\; G_1\, y_{t-1} \;+\; \Psi\, \varepsilon_t \;+\; \Pi\, \eta_t, \tag{67}$$

where ε_t collects the four structural innovations $(\varepsilon^{\lambda}, \varepsilon^{g}, \varepsilon^{s}, \varepsilon^{u^i})$ and η_t the four one-step-ahead forecast errors. Each structural row of G_0, G_1 corresponds to one labelled equation of the model section; the expectation variables $f_t^x \equiv \mathbb{E}_t x_{t+1}$ are tied to realizations by the forecast-error identities

$x_t = f^x_{t-1} + \eta^x_t$, the only rows touching Π, while the exogenous block contributes the four AR(1) rows $z_t = \rho_z z_{t-1} + \varepsilon^z_t$, the only rows touching Ψ.

The deterministic steady state is solved, not assigned: given the parameters, $Q^S = \beta/\bar{\pi}$, $Q^{L,f} = \beta/(\bar{\pi} - \beta\delta)$, $Q^L = (1+\omega)Q^{L,f}$, and $\tilde{\nu} = \frac{\bar{\pi}-\beta\delta}{\bar{\pi}} \cdot \frac{\omega}{1+\omega}$. The wedge is recovered from observables as $\omega \approx D \times \text{spread}$, with the Macaulay duration $D = 1/(1-\beta\delta/\bar{\pi})$ implied by $(\beta, \delta, \bar{\pi})$, so the calibrated wedge is tied to the calibration section rather than hard-coded. These objects supply the duration coefficients η, η_f, the Phillips slope κ, the value shares (w^v_S, w^v_L), and the backing shares $(R/v,\ s/v)$ that populate the linearized rows; the identity $s/v + R/v = 1 - \beta$ is checked at the calibration.

Equation (67) is solved for the bounded policy function $y_t = \Theta_1 y_{t-1} + \Theta_0 \varepsilon_t$ by an ordered generalized-Schur (QZ) decomposition. The routine factors the pencil (G_0, G_1) into upper-triangular (AA, BB) with unitary (Q, Z) and reports generalized eigenvalues $w_{ii} = AA_{ii}/BB_{ii}$; the dynamic eigenvalue governing stability is the reciprocal $\mu_{ii} = BB_{ii}/AA_{ii}$, and a root is explosive iff $|\mu_{ii}| > 1$. The decomposition is ordered so that stable roots occupy the leading block and explosive roots the trailing block, and the forecast errors are pinned by zeroing the explosive canonical directions. By the Blanchard–Kahn count a unique bounded solution exists iff the number of explosive roots equals the number of forecast errors, here four; at the baseline calibration the solver returns $eu = [1, 1]$, confirming determinacy.

Two variants share the builder. Model I (passive money, $\phi_\pi < 1$; active fiscal) takes the surplus as an exogenous AR(1). Model II (active money, $\phi_\pi = 1.5$; passive fiscal) replaces the surplus's own persistence with the debt-stabilizing rule $\hat{s}_t = \gamma_v \hat{v}_{t-1} + \hat{s}^{\text{exo}}_t$ on the single surplus row. The flexible-price benchmark is not a separate model: it is either variant run with the Calvo parameter sent to a small positive value $\theta \to 0^+$, which makes the Phillips curve (near-)vertical and returns the real allocation to its natural level.

```python
import numpy as np
from scipy.linalg import ordqz

THETA_FLEX = 1e-6
PHI_ACTIVE = 1.5


def calibrate():
    p = dict(
        beta=0.99, sigma=1.0, theta=0.75, varphi=0.0,
        delta=0.998, wL=0.353, sg=0.40, pi_bar=1.005,
        spread=0.0023,
        phi_pi=0.5, rho_lam=0.95, rho_g=0.97, rho_s=0.90, rho_ui=0.15,
        Vbar=1.0, gamma_v=10.0,
    )
    D_years = (1.0 / (1.0 - p['beta'] * p['delta'] / p['pi_bar'])) / 4.0
    p['omega'] = D_years * p['spread']
    return p


def steady(p):
    b, s, th, d = p['beta'], p['sigma'], p['theta'], p['delta']
    pi, om, wL = p['pi_bar'], p['omega'], p['wL']
    QS = b / pi
    QLf = b / (pi - b * d)
    QL = (1 + om) * QLf
    nu = (pi - b * d) / pi * om / (1 + om)
    eta_f = b * d / pi
    eta = d * QL / (1 + d * QL)
    kap = (1 - th) * (1 - b * th) / th
    bL = wL / QL
```

```python
    bS = (1 - wL) / QS
    v = bS + (1 + d * QL) * bL
    wSv = bS / v
    wLv = (1 + d * QL) * bL / v
    R = nu * QL * bL
    Rv = R / v
    sv = (1 - b) - Rv
    return dict(QS=QS, QLf=QLf, QL=QL, nu=nu, eta_f=eta_f, eta=eta, kap=kap,
                bL=bL, bS=bS, v=v, wSv=wSv, wLv=wLv, R=R, Rv=Rv, sv=sv)


V = ['c', 'y', 'pi', 'i', 'QS', 'QL', 'QLf', 'om', 'nu', 'R', 'v', 'bS', 'bL', 's',
     'lam', 'g', 'sx', 'ui',
     'fc', 'fpi', 'fQL', 'fQLf']
ix = {n: j for j, n in enumerate(V)}
N = len(V)
SH = {'e_lam': 0, 'e_g': 1, 'e_s': 2, 'e_ui': 3}
ET = {'c': 0, 'pi': 1, 'QL': 2, 'QLf': 3}


def build(p, model=1):
    pp = dict(p)
    if model == 2:
        pp['phi_pi'] = PHI_ACTIVE
    ss = steady(pp)
    b = pp['beta']; sig = pp['sigma']; sg = pp['sg']
    phi = pp['phi_pi']; gv = pp['gamma_v']
    nu, eta, etaf, kap = ss['nu'], ss['eta'], ss['eta_f'], ss['kap']
    wSv, wLv, Rv, om = ss['wSv'], ss['wLv'], ss['Rv'], pp['omega']
    sv = ss['sv']

    G0 = np.zeros((N, N))
    G1 = np.zeros((N, N))
    PSI = np.zeros((N, 4))
    PI = np.zeros((N, 4))

    def g0(r, n, x): G0[r, ix[n]] = x
    def g1(r, n, x): G1[r, ix[n]] = x

    g0(0, 'c', 1); g0(0, 'fc', -1); g0(0, 'i', 1 / sig); g0(0, 'fpi', -1 / sig)
    g0(1, 'pi', 1); g0(1, 'fpi', -b); g0(1, 'c', -kap * sig)
    g0(2, 'i', 1); g0(2, 'pi', -phi); g0(2, 'ui', -1)
    g0(3, 'QS', 1); g0(3, 'i', 1)
    g0(4, 'QLf', 1); g0(4, 'i', 1); g0(4, 'fQLf', -etaf)
    for n, x in [('QL', 1), ('nu', -1 / (1 - nu)), ('fc', sig), ('c', -sig), ('fpi', 1), ('fQL',
     -eta)]:
        g0(5, n, x)
    g0(6, 'om', 1); g0(6, 'QL', -(1 + om)); g0(6, 'QLf', (1 + om))
    g0(7, 'QL', 1); g0(7, 'bL', 1); g0(7, 'lam', -1)
    g0(8, 'R', 1); g0(8, 'nu', -1 / nu); g0(8, 'QL', -1); g0(8, 'bL', -1)
    g0(9, 'bL', 1); g0(9, 'bS', -1)
    g0(10, 'y', 1); g0(10, 'c', -(1 - sg)); g0(10, 'g', -sg)
    g0(11, 'v', 1); g0(11, 'pi', 1); g0(11, 'QL', -wLv * eta)
    g1(11, 'bS', wSv); g1(11, 'bL', wLv)
    for n, x in [('v', 1), ('s', -sv), ('R', -Rv), ('bS', -b * wSv), ('bL', -b * wLv),
                 ('fQL', -b * wLv * eta), ('i', b)]:
        g0(12, n, x)
    g0(13, 's', 1); g0(13, 'sx', -1)

    rho_z = {'lam': pp['rho_lam'], 'g': pp['rho_g'], 'sx': pp['rho_s'], 'ui': pp['rho_ui']}
    for r, z in zip(range(14, 18), ['lam', 'g', 'sx', 'ui']):
```

```
        g0(r, z, 1); g1(r, z, rho_z[z]); PSI[r, r - 14] = 1
    for r, (x, f) in zip(range(18, 22), [('c', 'fc'), ('pi', 'fpi'), ('QL', 'fQL'), ('QLf', 'fQLf')]):
        g0(r, x, 1); g1(r, f, 1); PI[r, ET[x]] = 1

    if model == 2:
        G1[16, ix['sx']] = 0.0
        G1[16, ix['v']] = gv

    return G0, G1, PSI, PI


def gensys(g0, g1, psi, pi, div=1.0):
    n = g0.shape[0]
    AA, BB, alpha, beta, Q, Z = ordqz(g0, g1, sort='ouc', output='complex')
    eigmod = np.abs(beta) / np.abs(alpha)
    nunstab = int(np.sum(eigmod > div + 1e-9))
    neta = pi.shape[1]
    nstab = n - nunstab

    Qh = Q.conj().T
    qpsi = Qh @ psi
    qpi = Qh @ pi
    qpsi2, qpi2 = qpsi[nstab:, :], qpi[nstab:, :]
    qpsi1, qpi1 = qpsi[:nstab, :], qpi[:nstab, :]

    if nunstab == neta:
        phi = -np.linalg.solve(qpi2, qpsi2); eu = [1, 1]
    elif nunstab < neta:
        phi = -np.linalg.pinv(qpi2) @ qpsi2; eu = [1, 0]
    else:
        phi = np.zeros((neta, psi.shape[1])); eu = [0, 0]

    a11, b11 = AA[:nstab, :nstab], BB[:nstab, :nstab]
    M11 = np.linalg.solve(a11, b11)
    Nn = np.linalg.solve(a11, qpsi1 + qpi1 @ phi)
    Z1 = Z[:, :nstab]
    THETA1 = np.real(Z1 @ M11 @ Z1.conj().T)
    THETA0 = np.real(Z1 @ Nn)
    return THETA1, THETA0, eu, np.sort(eigmod)[::-1]


def irf(THETA1, THETA0, shock, H):
    out = np.zeros((N, H + 1))
    e = np.zeros(4); e[SH[shock]] = 1.0
    y = THETA0 @ e
    out[:, 0] = y
    for h in range(1, H + 1):
        y = THETA1 @ y
        out[:, h] = y
    return out


def solve(p=None, model=1):
    if p is None:
        p = calibrate()
    G0, G1, PSI, PI = build(p, model=model)
    return gensys(G0, G1, PSI, PI)
```

11. Appendix 5: Proposition 1 in General Equilibrium

Proposition 1 gives the partial $\partial R/\partial\lambda = \tilde{\nu}\bar{V} > 0$, differentiating $R = \tilde{\nu}\,\lambda\,\bar{V}$ with the shadow value $\tilde{\nu}$ held fixed. Because $\tilde{\nu}$, Q^L and the quantities are endogenous, this appendix records how that response behaves in general equilibrium read off the solved linear policy function.

Holding the calibrated structural wedge ω fixed, $\tilde{\nu}$ is pinned by (32) and does not depend on λ; a change in λ moves the captive quantity $b^L = \lambda\bar{V}/Q^L$, not the shadow price. Thus $d\tilde{\nu}/d\lambda = 0$ and the level derivative equals the partial, $dR/d\lambda = \tilde{\nu}\bar{V}$ exactly. The "$\tilde{\nu}$ fixed" step is not an approximation but the steady-state truth.

Dynamics differ. From the coverage constraint ($\hat{Q}^L_t + \hat{b}^L_t = \hat{\lambda}_t$) and the rent definition, the rent's log-response is $\hat{R} = d\ln\tilde{\nu} + \hat{\lambda}$, so, normalising $\hat{\lambda}$ to the innovation, the impact elasticity decomposes as

$$\left.\frac{d\ln R}{d\ln\lambda}\right|_{\text{eq}} = \frac{\hat{R}_0}{\hat{\lambda}_0} = \underbrace{1}_{\text{direct}} + \underbrace{\frac{d\ln\tilde{\nu}}{d\ln\lambda}}_{\text{indirect}}. \tag{68}$$

This is the (log) elasticity, distinct from the level derivative $dR/d\lambda$ above; the partial sets the indirect term to zero, and the sign is preserved whenever the indirect term exceeds -1.

term	value
direct (partial, $\tilde{\nu}$ held fixed)	+1.000
indirect (GE feedback, $d\ln\tilde{\nu}/d\ln\lambda$)	+114.9
$\hat{R}_0/\hat{\lambda}_0$ (impact elasticity)	+115.9
determinacy flags eu	$[1,1]$

The indirect term is positive: the equilibrium feedback reinforces the partial rather than opposing it. Its sign follows from the mechanism: a tighter mandate forces additional captive holding against price-inelastic demand, raising Q^L relative to $Q^{L,f}$ and widening the wedge ($\hat{\omega}_0 > 0$), while $\tilde{\nu}$ is strictly increasing in ω ($d\tilde{\nu}/d\omega = \frac{\bar{\pi}-\beta\delta}{\bar{\pi}}(1+\omega)^{-2} > 0$). Its large magnitude reflects the tiny base $\tilde{\nu} \approx 5.6\times10^{-4}$; scaled to backing, the same response is $(R/v)\hat{R}_0 = +0.22\%$ of debt for a ten-percent tightening. What matters for the proposition is the sign, positive by a wide margin, as the sign-flip threshold (indirect $= -1$) is nowhere near.

The impact elasticity is strictly positive, and the system determinate ($eu = [1,1]$), at every point of the following sweeps: the wedge from its 23 bp floor to the 70 bp band; the long-bond decay $\delta \in [0.990, 0.999]$; the Taylor coefficient $\phi_\pi \in [0, 0.95]$ toward the passive-money determinacy boundary; and the regulatory persistence $\rho_\lambda \in [0.80, 0.98]$. The result also holds in the active-money, passive-fiscal regime. Proposition 1's sign therefore holds in general equilibrium throughout the binding regime: the dynamic feedback reinforces the partial rather than offsetting it.